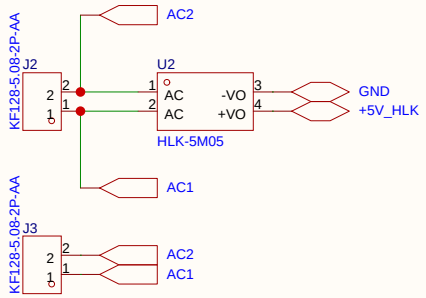
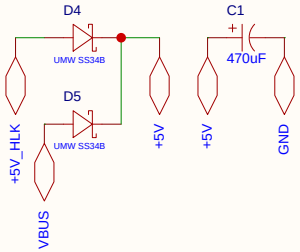
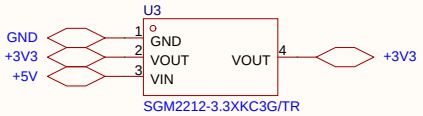
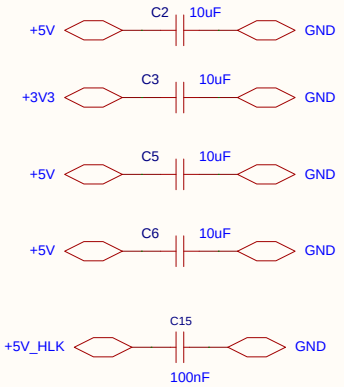
MAINS INPUT 230 V



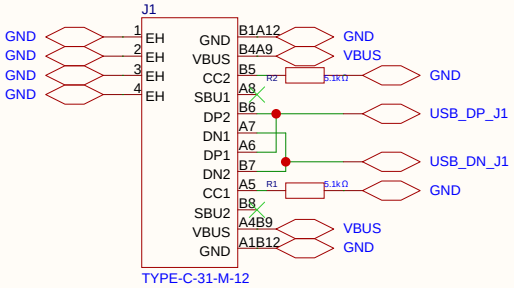
5 V / 3.3 V POWER

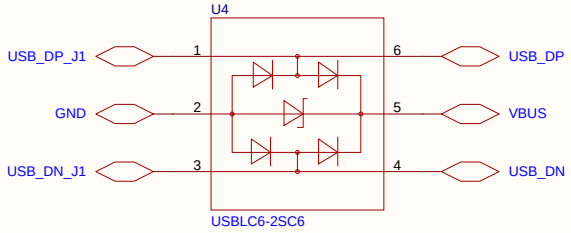


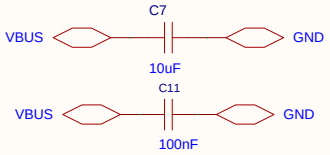




USB-C PROGRAMMING

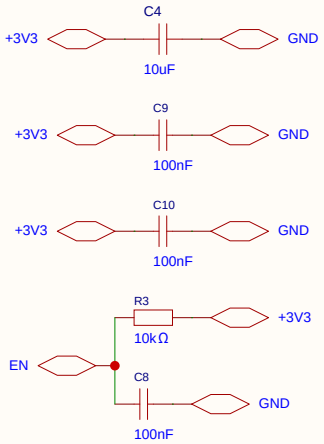


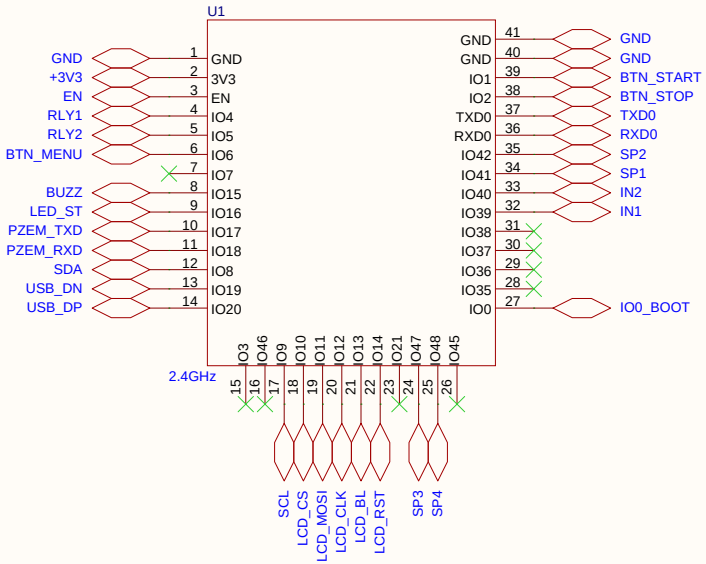


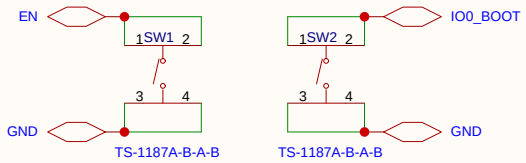


Schematic	Schematic1	Create at	2026-10-28
		Update at	2026-10-29
Board	Board1	Page	Power and USB
Drawn		BorePilot-Smart water pump	
Reviewed			
		Version	Size	Page 1 Total 5
EasyEDA		V1.0	A4	EasyEDA.com

ESP32-S3 CORE

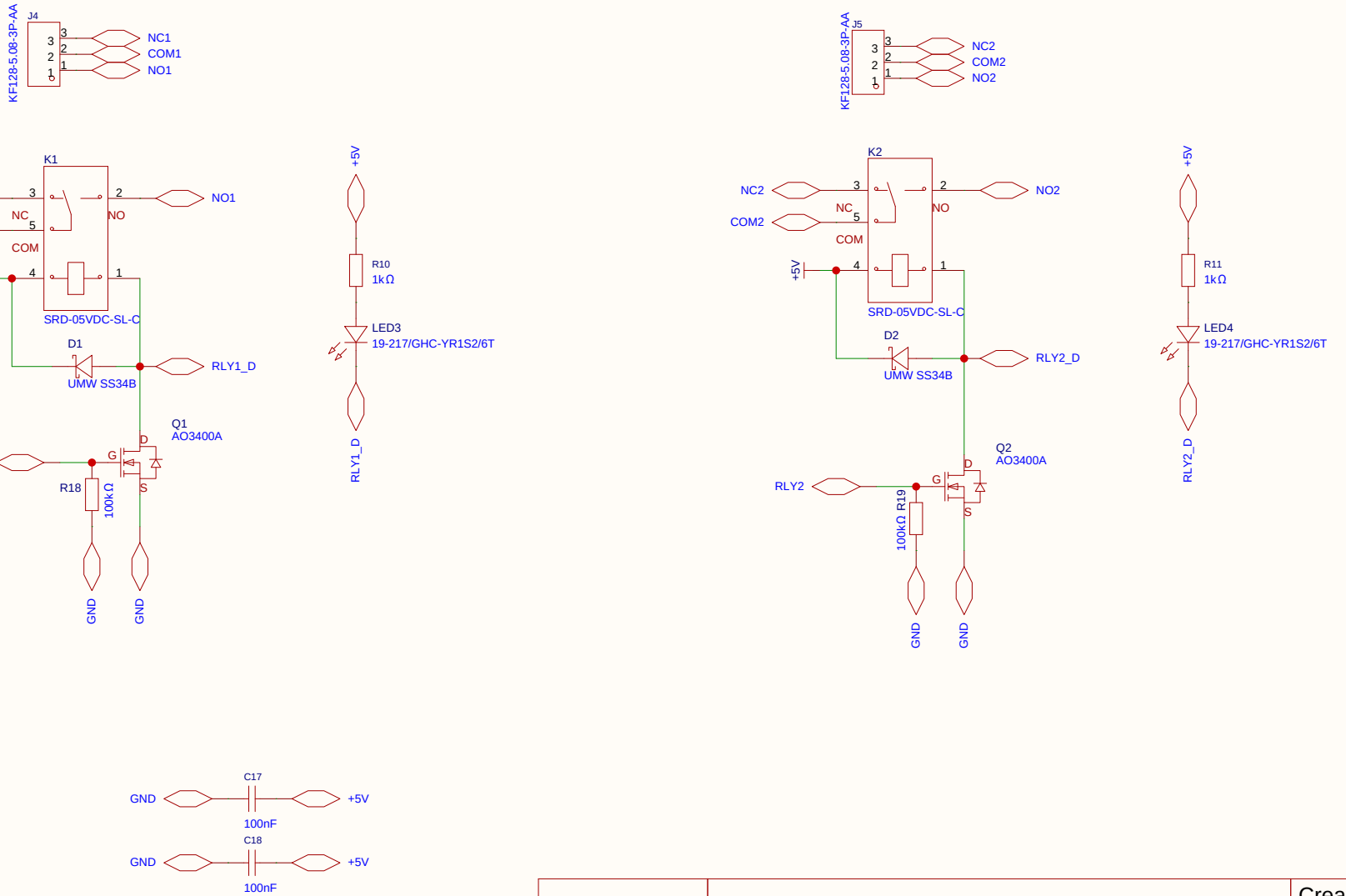






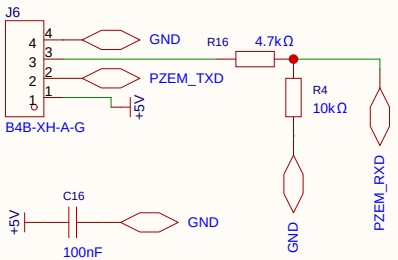
Schematic	Schematic1	Create at	2026-10-28
		Update at	2026-10-28
Board	Board1	Page	ESP32-S3 core
Drawn		BorePilot-Smart water pump	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 2 Total 5	
		EasyEDA.com	

Relays and mains contacts

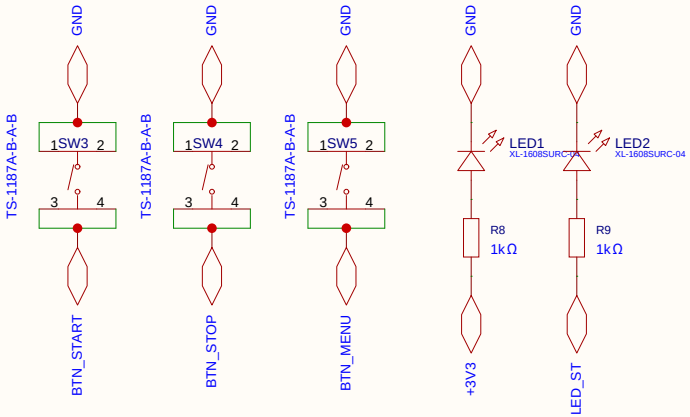


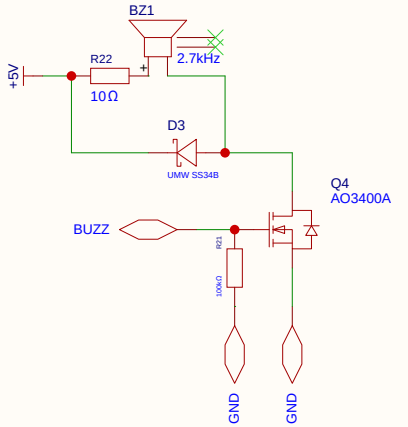
Schematic	Schematic1	Create at	2026-10-28
		Update at	2026-11-11
Board	Board1	Page	Relays and mains contacts
Drawn		BorePilot-Smart water pump	
Reviewed			
		Version	Size	Page 3 Total 5
EasyEDA		V1.0	A4	EasyEDA.com

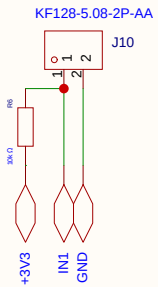
PZEM-004T LINK: header, divider on the module's TX line

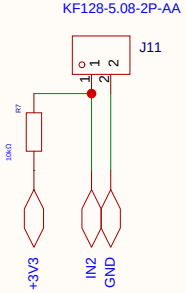


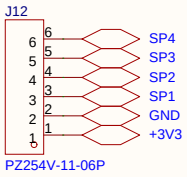
BUTTONS, LEDs, FLOAT INPUTS and OPTIONAL BUZZER

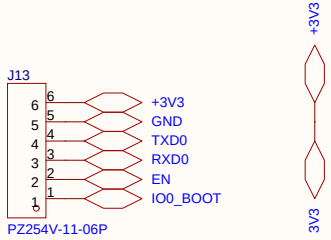






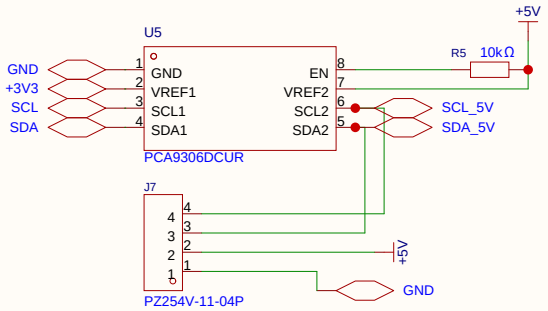




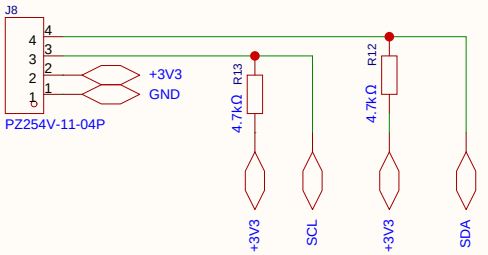


Schematic	Schematic1	Create at	2026-10-28
		Update at	2026-10-29
Board	Board1	Page	PZEM link and I/O
Drawn		BorePilot-Smart water pump	
Reviewed			
		Version	Size	Page 4 Total 5
EasyEDA		V1.0	A4	EasyEDA.com

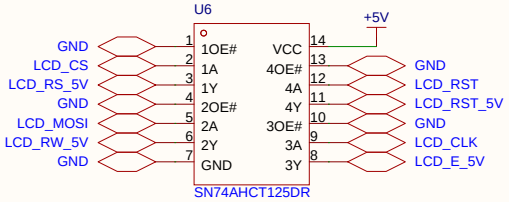
OPTION A: 16x2 I2C LCD through the PCA9306 level shifter

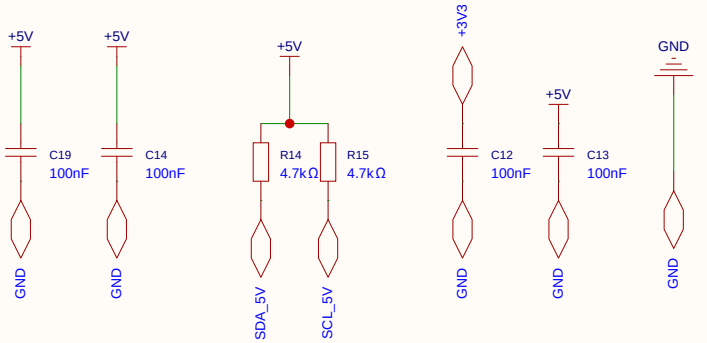


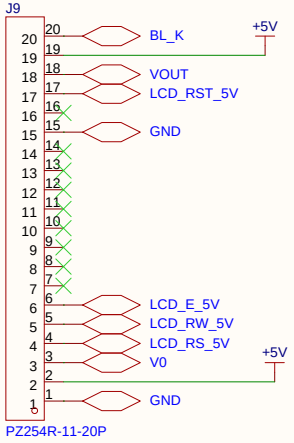
OPTION B: I2C OLED, 3.3 V direct

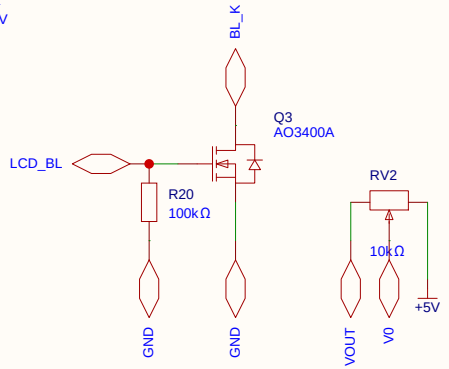


OPTION C: 12864 ST7920 through the 74AHCT125, backlight and contrast









Schematic	Schematic1	Create at	2026-10-28
		Update at	2026-10-28
Board	Board1	Page	Display options
Drawn		BorePilot-Smart water pump	
Reviewed			
		Version	Size
		V1.0	A4
		Page 5 Total 5	
		EasyEDA.com	